NeuroSky MindWave Mobile Android SDK

Developer Guide · v2.0.1

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1. Overview

The **NeuroSky MindWave Mobile Android SDK** is a modern Kotlin library that lets you read real-time EEG (electroencephalography) data from a NeuroSky MindWave Mobile headset on Android 6.0 (API 23) or later — with zero dependency on NeuroSky's legacy ThinkGear Connector (TGC) software.

Why this SDK exists

The official NeuroSky Android SDK requires TGC running on a paired PC, communicating over TCP, and is built on a legacy Java API that does not work well with modern Android development patterns (Jetpack, Coroutines, Flow, ViewModel).

This SDK eliminates TGC entirely by communicating directly with the MindWave Mobile hardware via Android's Bluetooth stack. It is built on Kotlin Coroutines and Flow — the standard async primitives in modern Android.

Key features

Feature	Description
No TGC dependency	Communicates with hardware directly via Android Bluetooth APIs
BLE + BT Classic	BLE by default; BT Classic available for noisy RF environments
Kotlin Coroutines & Flow	`Flow<BrainWaveData>` — integrates naturally with Jetpack lifecycle
Built-in Simulator	Full data simulation without any hardware
JitPack distribution	One-line Gradle dependency, no local setup

Android 6.0+ (API 23+)	Wide device coverage

What you can measure

The MindWave Mobile headset contains a single dry electrode on the forehead (FP1 position) and a reference clip on the ear. From this single channel, the ThinkGear ASIC chip on board computes:

- **Raw EEG waveform** — 512 samples/sec, signed 16-bit values
- **8 frequency band powers** — Delta, Theta, Alpha (Low/High), Beta (Low/High), Gamma (Low/Mid)
- **eSense™ Attention** — NeuroSky's proprietary attention index (0~100)
- **eSense™ Meditation** — NeuroSky's proprietary relaxation index (0~100)
- **Eye blink detection** — intensity 0~255
- **Signal quality** — 0 (perfect contact) to 200 (no signal)

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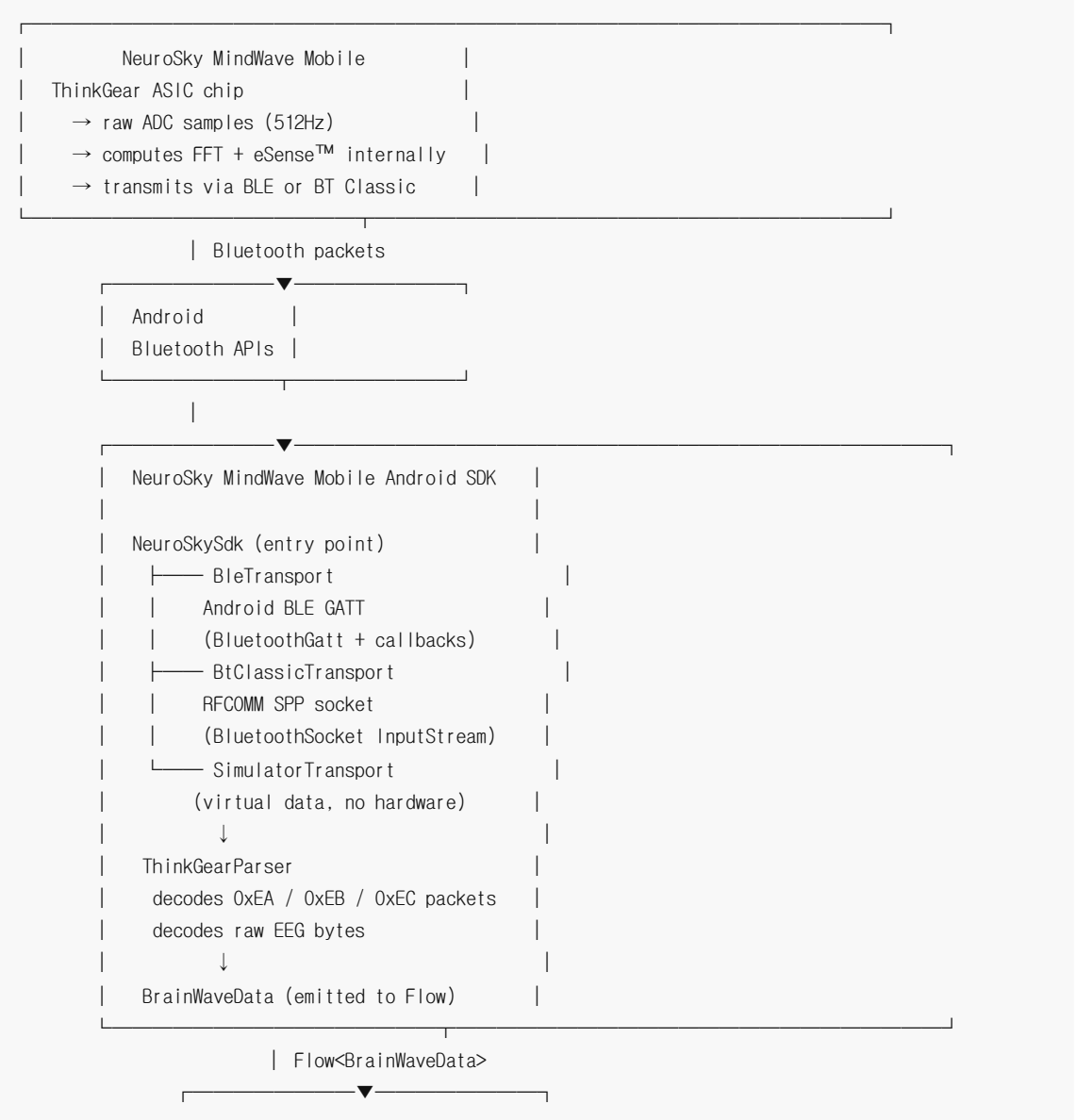
2. How It Works — Architecture



```
|  Your App        |
|  .collect { }    |
└──────────────────────────────────┘
```

BLE vs BT Classic — internal differences

BLE (Bluetooth Low Energy) path: The MindWave Mobile exposes three BLE GATT characteristics. The SDK subscribes to notifications on the eSense and RawEEG characteristics, then writes the handshake command byte to start data flow. No Android pairing is required.

BT Classic (RFCOMM SPP) path: The MindWave Mobile emulates a serial port (SPP UUID `00001101-...`). The SDK opens a `BluetoothSocket` and reads a continuous byte stream. `ThinkGearParser` synchronizes on the `0xAA 0xAA` sync header. The device must be paired in Android Bluetooth settings first.

Both paths produce identical `BrainWaveData` output through the same `dataFlow`.

BLE default — no automatic fallback

`NeuroSkySdk.connect()` uses BLE by default. To use BT Classic instead, pass `TransportType.BT_CLASSIC` to `connect()`. Both transports produce the same `dataFlow` output.

> *`NeuroSkySdk` does **not** attempt BLE first and then fall back to BT Classic automatically. The transport you pass to `connect()` is the only transport used. If you need fallback logic, implement it yourself in the caller.*

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3. Requirements

Device requirements

Component	Minimum
Android OS	Android 6.0 (API level 23)
Kotlin	1.9+
Coroutines	`kotlinx-coroutines-android` 1.7+
Bluetooth	BLE adapter (for BLE mode) or Classic BT (for BT Classic mode)
Pairing	Not required for BLE; required for BT Classic

Supported headset

This SDK is designed and tested for the **NeuroSky MindWave Mobile 2**. Both BLE and BT Classic modes are supported.

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4. Installation

Step 1 — Add JitPack to your repository list

In `settings.gradle.kts`:

```
dependencyResolutionManagement {
    repositoriesMode.set(RepositoriesMode.FAIL_ON_PROJECT_REPOS)
```

```
    repositories {
        google()
        mavenCentral()
        maven { url = uri("https://jitpack.io") }  // add this
    }
}
```

Step 2 — Add the dependency

In `app/build.gradle.kts`:

```
dependencies {
    implementation("com.github.nsk-bci:mindwave-sdk-android:v2.0.1")
}
```

Step 3 — Sync and verify

```
Android Studio → File → Sync Project with Gradle Files
```

Or via terminal:

```
./gradlew assembleDebug
```

If the build succeeds without errors, the SDK is correctly installed.

> ***Note:*** *JitPack builds the library on first request. The very first* `./gradlew` *run may take 30–60 seconds while JitPack fetches and compiles the source. Subsequent builds use the cached artifact.*

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5. Permissions

Android requires Bluetooth permissions in `AndroidManifest.xml`. The required permissions differ by Android version.

AndroidManifest.xml

```
<!-- Android 12+ (API 31+) -->
<uses-permission android:name="android.permission.BLUETOOTH_SCAN"
    android:usesPermissionFlags="neverForLocation" />
<uses-permission android:name="android.permission.BLUETOOTH_CONNECT" />

<!-- Android 6.0-11 (API 23-30) -->
<uses-permission android:name="android.permission.BLUETOOTH" />
<uses-permission android:name="android.permission.BLUETOOTH_ADMIN" />
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
```

> *Adding all of the above covers all Android versions from 6.0 to 14+. The OS ignores permissions not applicable to the current version.*

Runtime permission request (Android 12+)

On Android 12 and above, `BLUETOOTH_SCAN` and `BLUETOOTH_CONNECT` are runtime permissions — the user must grant them at runtime, not just at install time.

```
// In your Activity or Fragment
private val bluetoothPermissionLauncher = registerForActivityResult(
    ActivityResultContracts.RequestMultiplePermissions()
) { permissions ->
    val allGranted = permissions.values.all { it }
    if (allGranted) {
        startEegSession()
    } else {
        showPermissionDeniedMessage()
    }
}

private fun requestBluetoothPermissions() {
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
        bluetoothPermissionLauncher.launch(
            arrayOf(
                Manifest.permission.BLUETOOTH_SCAN,
                Manifest.permission.BLUETOOTH_CONNECT
            )
        )
    } else {
        // Android 11 and below: request location permission
        bluetoothPermissionLauncher.launch(
            arrayOf(Manifest.permission.ACCESS_FINE_LOCATION)
        )
    }
}
```

Why location permission is needed (Android 6–11)

On Android versions before 12, BLE scanning is considered a location-revealing operation because BLE beacons can be used for indoor positioning. The OS therefore requires `ACCESS_FINE_LOCATION` to perform BLE scans — even if your app has nothing to do with location. This requirement was lifted in Android 12 with the introduction of `BLUETOOTH_SCAN`.

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6. Quick Start

The following example is a complete minimal implementation inside an Activity using `lifecycleScope`:

```
import com.neurosky.sdk.NeuroSkySdk
import com.neurosky.sdk.NeuroSkyCommand
import com.neurosky.sdk.model.SignalQuality
import com.neurosky.sdk.transport.ConnectionState

class MainActivity : AppCompatActivity() {
```

```
    private lateinit var sdk: NeuroSkySdk

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        // Step 1: Create SDK instance (requires Context)
        sdk = NeuroSkySdk(this)

        // Step 2: Observe connection state
        lifecycleScope.launch {
            sdk.connectionState.collect { state ->
                when (state) {
                    ConnectionState.SCANNING    -> updateStatus("Scanning...")
                    ConnectionState.CONNECTING  -> updateStatus("Connecting...")
                    ConnectionState.CONNECTED   -> updateStatus("Connected")
                    ConnectionState.ERROR       -> updateStatus("Connection failed")
                    ConnectionState.DISCONNECTED -> updateStatus("Disconnected")
                }
            }
        }

        // Step 3: Connect and stream data
        lifecycleScope.launch {
            // Connect using device name (BLE by default)
            sdk.connect("MindWave Mobile")

            // Set notch filter for your region
            sdk.sendCommand(NeuroSkyCommand.NOTCH_60HZ)  // Korea/USA
            // sdk.sendCommand(NeuroSkyCommand.NOTCH_50HZ)  // Europe/China

            // Collect EEG data stream
            sdk.dataFlow.collect { data ->
                if (data.signalQuality == SignalQuality.NO_SIGNAL) {
                    updateStatus("No signal - adjust headset")
                    return@collect
                }

                attentionBar.progress  = data.attention
                meditationBar.progress = data.meditation
                signalText.text        = data.signalQuality.name
            }
        }
    }

    override fun onDestroy() {
        super.onDestroy()
        lifecycleScope.launch { sdk.disconnect() }
    }
}
```

Connecting by MAC address vs device name

`connect()` accepts either a device name or a Bluetooth MAC address:

```
// By device name (BLE scan required — slower, ~5 sec)
sdk.connect("MindWave Mobile")

// By MAC address (direct connect — faster, recommended for production)
sdk.connect("AA:BB:CC:DD:EE:FF")
```

Using the MAC address is faster and more reliable in production. Use device name during prototyping when you don't yet know the MAC.

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7. EEG Data Model

`dataFlow` emits a `BrainWaveData` object for each packet received from the MindWave Mobile headset.

```
data class BrainWaveData(
    val timestamp:   Long,          // System.currentTimeMillis() at receipt
    val poorSignal:  Int,           // 0 = perfect, 200 = no contact
    val attention:   Int,           // 0~100, eSense™ attention
    val meditation:  Int,           // 0~100, eSense™ meditation
    val delta:       Int,           // 0.5~2.75 Hz band power
    val theta:       Int,           // 3.5~6.75 Hz band power
    val lowAlpha:    Int,           // 7.5~9.25 Hz band power
    val highAlpha:   Int,           // 10~11.75 Hz band power
    val lowBeta:     Int,           // 13~16.75 Hz band power
    val highBeta:    Int,           // 18~29.75 Hz band power
    val lowGamma:    Int,           // 31~39.75 Hz band power
    val midGamma:    Int,           // 41~49.75 Hz band power
    val rawEeg:      List<Int>,     // 10 samples/packet, signed 16-bit, 512Hz
    val eyeBlink:    Int,           // 0 = no blink, 1~255 = blink intensity
) {
    val signalQuality: SignalQuality  // derived from poorSignal
}
```

Data update rates

Field(s)	Update rate	Notes
`poorSignal`	~1 Hz	Every packet
`attention`, `meditation`	~1 Hz	eSense™ computed once per second
`delta` through `midGamma`	~1 Hz	FFT computed once per second
`rawEeg`	512 Hz	10 samples per packet, ~51 packets/sec
`eyeBlink`	Event-driven	Only non-zero when blink detected

> *When `rawEeg` packets arrive, `attention`, `meditation`, and frequency band fields are `0` in that object — they only appear in the eSense packet which arrives separately. Filter by checking which fields are non-zero.*

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8. EEG Frequency Bands Explained

The ThinkGear chip performs a Fast Fourier Transform (FFT) on the raw EEG and outputs power in 8 frequency bands.

Important: values are relative, not absolute

The frequency band values are **relative power** in arbitrary units — not calibrated to physical units (μV²/Hz). This means:

- You **cannot** compare values across different sessions or individuals in absolute terms
- You **can** compare values within a single session — e.g., "Delta rose 40% after eyes closed"
- **Ratios** between bands are more meaningful than raw values

Band reference table

Field	Band	Frequency	Typical mental states
`delta`	δ Delta	0.5~2.75 Hz	Deep sleep, healing. High delta while awake = fatigue or poor signal.
`theta`	θ Theta	3.5~6.75 Hz	Drowsiness, daydreaming, creativity, REM sleep, deep meditation.
`lowAlpha`	α Low	7.5~9.25 Hz	Relaxed, unfocused, calm. Increases with eyes closed.
`highAlpha`	α High	10~11.75 Hz	Eyes-closed rest. Suppressed by active visual attention.
`lowBeta`	β Low	13~16.75 Hz	Active focus, alert thinking. The "work" band.
`highBeta`	β High	18~29.75 Hz	High arousal, stress, anxiety, intense cognition.
`lowGamma`	γ Low	31~39.75 Hz	Higher cognition, cross-modal perception, binding.
`midGamma`	γ Mid	41~49.75 Hz	Intense concentration. Elevated in expert meditators.

eSense™ Attention and Meditation

These are NeuroSky's **proprietary processed values** computed by the ThinkGear chip itself. The SDK receives them pre-computed.

Attention (0~100) — mental focus or concentration:

Range	Meaning
0	Not yet computed (startup, or no signal)

1~40	Low — distracted, relaxed, wandering mind
40~60	Neutral baseline
60~80	Moderate focus — engaged in task
80~100	High focus — strong active concentration

Meditation (0~100) — mental calmness or relaxation:

Range	Meaning
0	Not yet computed (startup, or no signal)
1~40	Low — active thinking, stress
40~60	Neutral baseline
60~80	Moderate relaxation
80~100	Deep calm — strong meditative state

> *eSense values require 10~20 seconds to stabilize after the headset is put on. Values of 0 at startup are normal.*

Raw EEG

When Raw EEG is enabled, `rawEeg` contains 10 signed 16-bit samples per packet at 512 Hz:

```
// Enable raw EEG after connecting
sdk.sendCommand(NeuroSkyCommand.START_RAW_EEG)

sdk.dataFlow.collect { data ->
    data.rawEeg.forEach { sample ->
        // Each sample: -32768 to +32767
        // 512 Hz, 10 samples per packet
        plotSample(sample)
    }
}
```

Raw EEG is useful for custom FFT analysis, artifact detection, or research. It is **disabled by default** to reduce Bluetooth bandwidth.

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9. Signal Quality

Signal quality is the most critical factor for usable data. Always check it before using attention, meditation, or band values.

PoorSignal values

Value	`signalQuality`	Reliability	Action
0	GOOD	Excellent	Use all data freely

1~50	`FAIR`	Acceptable	Minor noise, eSense still valid
51~199	`POOR`	Unreliable	Prompt user to adjust headset
200	`NO_SIGNAL`	No data	Headset not worn

Recommended check pattern

```
sdk.dataFlow.collect { data ->
    when (data.signalQuality) {
        SignalQuality.NO_SIGNAL -> {
            showMessage("Please put on the MindWave Mobile headset.")
            return@collect
        }
        SignalQuality.POOR -> {
            showMessage("Weak signal (${data.poorSignal}). Adjust the headset.")
            // Optionally still log, but don't drive UI decisions
            return@collect
        }
        SignalQuality.FAIR,
        SignalQuality.GOOD -> {
            // Data is usable — update UI
            updateAttentionUi(data.attention)
            updateMeditationUi(data.meditation)
        }
    }
}
```

Tips for improving signal quality

1. **Moisten the sensor** — a small drop of water on the forehead sensor significantly improves conductance
2. **Clean the forehead** — remove sunscreen, makeup, or sweat residue
3. **Adjust headset position** — center the sensor on FP1 (above the left eyebrow)
4. **Check the ear clip** — must make firm contact with the earlobe
5. **Wait 20~30 seconds** — after putting on the headset for signal to stabilize
6. **Avoid strong muscle movement** — clenching the jaw creates EMG artifact that looks like a signal-quality drop

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10. Commands

After connecting, send control commands to configure the MindWave Mobile headset's behavior.

Notch filter

EEG signals are extremely low amplitude and easily contaminated by AC power-line noise. Send the notch filter command immediately after connecting:

Region	Grid frequency	Command
Korea	60 Hz	`NeuroSkyCommand.NOTCH_60HZ`

USA / Canada	60 Hz	`NeuroSkyCommand.NOTCH_60HZ`
Europe	50 Hz	`NeuroSkyCommand.NOTCH_50HZ`
China	50 Hz	`NeuroSkyCommand.NOTCH_50HZ`
Australia / UK	50 Hz	`NeuroSkyCommand.NOTCH_50HZ`

```
sdk.connect("MindWave Mobile")
sdk.sendCommand(NeuroSkyCommand.NOTCH_60HZ)   // Korea/USA
// sdk.sendCommand(NeuroSkyCommand.NOTCH_50HZ)  // Europe/China
```

Without the notch filter, you will likely see a large 50Hz or 60Hz artifact in raw EEG and elevated Beta band values.

Raw EEG streaming

```
// Enable raw EEG (disabled by default)
sdk.sendCommand(NeuroSkyCommand.START_RAW_EEG)

// Disable when no longer needed
sdk.sendCommand(NeuroSkyCommand.STOP_RAW_EEG)
```

eSense control

```
// Disable eSense if only raw EEG is needed
sdk.sendCommand(NeuroSkyCommand.STOP_ESENSE)

// Re-enable
sdk.sendCommand(NeuroSkyCommand.START_ESENSE)
```

All commands

Constant	Byte	Description
`NeuroSkyCommand.NOTCH_60HZ`	`0x1C`	Notch filter at 60 Hz
`NeuroSkyCommand.NOTCH_50HZ`	`0x1B`	Notch filter at 50 Hz
`NeuroSkyCommand.START_RAW_EEG`	`0x15`	Enable raw EEG stream
`NeuroSkyCommand.STOP_RAW_EEG`	`0x16`	Disable raw EEG stream
`NeuroSkyCommand.START_ESENSE`	`0x17`	Enable eSense output
`NeuroSkyCommand.STOP_ESENSE`	`0x18`	Disable eSense output

---

11. Simulator — Develop Without Hardware

`SimulatorTransport` generates synthetic EEG data without any MindWave Mobile hardware. It implements `Transport`, so your application code remains unchanged between development and production.

Why use the Simulator

- **No hardware required** — build and test UI, data flows, and business logic immediately
- **Predictable data** — use `FOCUSED` mode to always generate high-attention values for UI testing
- **Edge case testing** — `POOR_SIGNAL` mode tests your error-handling and reconnect logic
- **CI/CD pipelines** — run Espresso or unit tests on build servers without Bluetooth hardware

Basic usage

```kotlin
import com.neurosky.sdk.simulator.SimulatorTransport  // package: simulator, NOT transport

val simulator = SimulatorTransport()
simulator.setMode(SimulatorTransport.Mode.FOCUSED)

lifecycleScope.launch {
    simulator.connect("simulator")  // any string accepted; CONNECTED after ~500 ms

    simulator.dataFlow.collect { data ->
        Log.d("SIM", "Attention: ${data.attention}, Meditation: ${data.meditation}")
    }
}
```

> *`stateFlow` **vs** `connectionState`: `SimulatorTransport` (and all `Transport` implementations) expose `stateFlow: Flow<ConnectionState>` from the `Transport` interface. `connectionState: StateFlow<ConnectionState>` is a property of `NeuroSkySdk` only — it does **not** exist on `SimulatorTransport` or the `Transport` interface directly.*
>
> ```kotlin
> // Wrong — connectionState does not exist on SimulatorTransport
> simulator.connectionState.collect { }   // compile error
>
> // Correct — use stateFlow (Transport interface)
> simulator.stateFlow.collect { state -> /* CONNECTED, DISCONNECTED, … */ }
> ```

```
### Simulator modes

| Mode | Attention | Meditation | PoorSignal | Use case |
|---|---|---|---|---|
| `RANDOM` | 0~100 (random) | 0~100 (random) | 0 | General integration testing |
| `FOCUSED` | 70~100 | 40~60 | 0 | High-attention UI testing |
| `RELAXED` | 20~50 | 70~100 | 0 | High-meditation UI testing |
| `POOR_SIGNAL` | 0 | 0 | 150~200 | Signal loss and error handling |

### Switching modes at runtime

```kotlin
val simulator = SimulatorTransport()
```
```

```
simulator.setMode(SimulatorTransport.Mode.POOR_SIGNAL)

lifecycleScope.launch {
    simulator.connect("simulator")

    // Simulate signal recovery after 5 seconds
    delay(5000)
    simulator.setMode(SimulatorTransport.Mode.FOCUSED)

    simulator.dataFlow.collect { data -> /* ... */ }
}
```

Using in ViewModel with dependency injection

```
// Swap between simulator and real SDK via DI or build flavor
class EegViewModel(
    private val transport: Transport  // injected: NeuroSkySdk or SimulatorTransport
) : ViewModel() {

    val data: Flow<BrainWaveData> = transport.dataFlow
    val state: Flow<ConnectionState> = transport.stateFlow

    fun connect(address: String) {
        viewModelScope.launch {
            transport.connect(address)
        }
    }
}
```

---

12. Error Handling & Reconnection

Connection errors

`connect()` is a suspend function — wrap in try/catch for production use:

```
lifecycleScope.launch {
    try {
        sdk.connect("MindWave Mobile")
    } catch (e: SecurityException) {
        // Bluetooth permission not granted
        showError("Bluetooth permission required.")
    } catch (e: Exception) {
        // BT adapter off, device not found, etc.
        showError("Connection failed: ${e.message}")
    }
}
```

Auto-reconnect loop

`dataFlow` completes (collection ends) when the connection drops. Wrap in a retry loop for production:

```
lifecycleScope.launch {
    while (isActive) {  // isActive = CoroutineScope is still alive
        try {
            sdk.connect("MindWave Mobile")
            sdk.sendCommand(NeuroSkyCommand.NOTCH_60HZ)

            sdk.dataFlow.collect { data ->
                processData(data)
            }

            // dataFlow completed — headset turned off
            updateStatus("Disconnected. Reconnecting...")

        } catch (e: CancellationException) {
            break  // ViewModel/lifecycle scope cancelled — exit cleanly
        } catch (e: Exception) {
            updateStatus("Error: ${e.message}. Retrying in 3s...")
        }

        delay(3000)  // wait before retry
    }
}
```

Handling NO_SIGNAL without disconnecting

The Bluetooth connection can remain active while the electrode is not touching the skin. In this case, `signalQuality` becomes `NO_SIGNAL` but `dataFlow` keeps emitting. Handle this in your collector:

```
sdk.dataFlow.collect { data ->
    if (data.signalQuality == SignalQuality.NO_SIGNAL) {
        updateUi(connected = true, signalOk = false)
        return@collect
    }
    updateUi(connected = true, signalOk = true)
    processData(data)
}
```

---

13. Advanced Patterns

ViewModel + StateFlow for Compose or XML UI

```
class EegViewModel(application: Application) : AndroidViewModel(application) {

    private val sdk = NeuroSkySdk(application)

    private val _attention  = MutableStateFlow(0)
    private val _meditation = MutableStateFlow(0)
    private val _signal     = MutableStateFlow(SignalQuality.NO_SIGNAL)
```

```
    val attention:  StateFlow<Int>           = _attention.asStateFlow()
    val meditation: StateFlow<Int>           = _meditation.asStateFlow()
    val signal:     StateFlow<SignalQuality> = _signal.asStateFlow()
    val state:      StateFlow<ConnectionState> = sdk.connectionState

    init {
        viewModelScope.launch {
            sdk.connect("MindWave Mobile")
            sdk.sendCommand(NeuroSkyCommand.NOTCH_60HZ)

            sdk.dataFlow.collect { data ->
                _signal.value     = data.signalQuality
                _attention.value  = data.attention
                _meditation.value = data.meditation
            }
        }
    }

    override fun onCleared() {
        viewModelScope.launch { sdk.disconnect() }
    }
}
```

Jetpack Compose integration

```
@Composable
fun EegScreen(viewModel: EegViewModel = viewModel()) {
    val attention  by viewModel.attention.collectAsState()
    val meditation by viewModel.meditation.collectAsState()
    val signal     by viewModel.signal.collectAsState()

    Column(modifier = Modifier.padding(16.dp)) {
        Text("Signal: $signal")
        LinearProgressIndicator(progress = attention / 100f)
        Text("Attention: $attention")
        LinearProgressIndicator(progress = meditation / 100f)
        Text("Meditation: $meditation")
    }
}
```

Buffering 1 second of raw EEG for custom analysis

```
// Raw EEG arrives as 10 samples per packet.
// Accumulate 512 samples for 1 second of data at 512 Hz.

val buffer = mutableListOf<Int>()

sdk.sendCommand(NeuroSkyCommand.START_RAW_EEG)
```

```
sdk.dataFlow.collect { data ->
    if (data.rawEeg.isEmpty()) return@collect

    buffer.addAll(data.rawEeg)

    if (buffer.size >= 512) {
        val oneSecond = buffer.take(512).toIntArray()
        buffer.subList(0, 512).clear()

        // Run your own FFT or signal processing
        val spectrum = MyFft.compute(oneSecond, sampleRate = 512)
        displaySpectrum(spectrum)
    }
}
```

Recording session data to a file

```
val filename = "eeg_${System.currentTimeMillis()}.csv"
val file = File(getExternalFilesDir(null), filename)

file.bufferedWriter().use { writer ->
    writer.write("timestamp,attention,meditation,poor_signal,delta,theta," +
                 "low_alpha,high_alpha,low_beta,high_beta,low_gamma,mid_gammaWn")

    sdk.dataFlow
        .filter { it.attention > 0 || it.meditation > 0 }  // skip raw-only packets
        .collect { d ->
            writer.write("${d.timestamp},${d.attention},${d.meditation},${d.poorSignal}," +
                         "${d.delta},${d.theta},${d.lowAlpha},${d.highAlpha}," +
                         "${d.lowBeta},${d.highBeta},${d.lowGamma},${d.midGamma}Wn")
        }
}
```

Background service with foreground notification

For applications that need to stream EEG data while the app is in the background:

```
class EegForegroundService : Service() {

    private val sdk by lazy { NeuroSkySdk(this) }
    private val scope = CoroutineScope(SupervisorJob() + Dispatchers.IO)

    override fun onStartCommand(intent: Intent?, flags: Int, startId: Int): Int {
        startForeground(NOTIFICATION_ID, buildNotification())

        scope.launch {
            sdk.connect("MindWave Mobile")
            sdk.dataFlow.collect { data ->
                broadcastData(data)  // send to Activity via LocalBroadcastManager
            }
```

```
        }

        return START_STICKY
    }

    override fun onDestroy() {
        scope.cancel()
        super.onDestroy()
    }
}
```

---

14. Troubleshooting

Connection issues

Symptom	Likely cause	Solution
`SecurityException` on `connect()`	Bluetooth permission not granted at runtime	Request `BLUETOOTH_SCAN` + `BLUETOOTH_CONNECT` (API 31+) or `ACCESS_FINE_LOCATION` (API 23–30)
`connect()` hangs indefinitely	BLE scan timeout	Use `findDeviceAddress()` first to resolve the MAC address, then pass the MAC to `connect()` for faster connection.
BT Classic connect fails	Device not paired	Open Android Settings → Bluetooth → pair "MindWave Mobile" manually
Connection drops after a few minutes	Android BLE background scan kill	Use a Foreground Service to prevent the OS from killing the connection
`dataFlow` stops emitting after screen off	Background execution limit	Same — use a Foreground Service

Signal quality issues

Symptom	Likely cause	Solution
`NO_SIGNAL` immediately after connecting	Electrode not touching forehead	Adjust headset and press sensor firmly to skin
`POOR` signal persists after 30 seconds	Dry skin or dirty sensor	Wet the sensor tip with water; clean forehead
`attention` and `meditation` always 0	eSense needs warm-up	Wait 20–30 seconds after achieving `GOOD` signal
50Hz or 60Hz spike in raw EEG	No notch filter	Call `sendCommand(NeuroSkyCommand.NOTCH_60HZ)` after connecting
`rawEeg` always empty	`START_RAW_EEG` not called	Call `sendCommand(NeuroSkyCommand.START_RAW_EEG)` after connecting

Build issues

Symptom	Likely cause	Solution
`Could not resolve com.github.nsk-bci:mindwave-sdk-android`	JitPack not in repository list	Add `maven { url = uri("https://jitpack.io") }` to `settings.gradle.kts`
`Unresolved reference: NeuroSkySdk`	Missing `import`	Add `import com.neurosky.sdk.NeuroSkySdk`
`Unresolved reference: SimulatorTransport`	Wrong import	Use `import com.neurosky.sdk.simulator.SimulatorTransport` (package is `simulator`, not `transport`)
`Unresolved reference: TransportMode`	Old doc typo	The correct enum is `TransportType`. Use `TransportType.BLE` / `TransportType.BT_CLASSIC`
`Unresolved reference: connectionState` on simulator	Wrong API	`connectionState` belongs to `NeuroSkySdk`. Use `simulator.stateFlow` instead
`Unresolved reference: alphaLow` / `betaHigh`	Wrong field name order	Field names are `lowAlpha`, `highAlpha`, `lowBeta`, `highBeta`, `lowGamma`, `midGamma` — modifier first
First build very slow	JitPack building from source	Normal — only happens once; subsequent builds use cache

Data issues

Symptom	Likely cause	Solution
`dataFlow` emits nothing even though connected	`dataFlow` captured before `connect()`	Move `sdk.dataFlow.collect` to inside the same coroutine after `connect()` returns
BLE connected but data always 0, NO_SIGNAL forever	Notch filter not set, or eSense not started	Call `sendCommand(NeuroSkyCommand.NOTCH_60HZ)` after connecting
Connection takes 5+ seconds	BLE scan by device name	Use `findDeviceAddress()` first, cache the MAC, then pass MAC to `connect()`

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15. Testing

The SDK ships with a unit test suite for `ThinkGearParser` — the packet parser that runs identically on both BLE and BT Classic transports. These tests require no hardware or Bluetooth adapter and run on the JVM directly.

Running the tests

```
./gradlew :sdk:test
```

Test results are written to:

```
sdk/build/reports/tests/testDebugUnitTest/index.html
```

What is covered

Test	Description
`parseEsense_0xEA`	Attention, Meditation, PoorSignal extraction
`parseEsense_0xEA_tooShort`	Short packet → returns null
`parseEsense_0xEB`	Delta, Theta, LowAlpha, HighAlpha extraction
`parseEsense_0xEC`	LowBeta, HighBeta, LowGamma, MidGamma extraction
`parseRawEeg_returns10Samples`	20-byte raw EEG → 10 signed int samples
`parseRawEeg_signedConversion`	Values > 32768 converted to negative
`parseRawEeg_tooShort`	Short packet → returns null
`parse_unknownUuid`	Unknown UUID → returns null
`parseByte_validPacket`	BT Classic serial packet — Attention/Meditation
`parseByte_invalidChecksum`	Wrong checksum → returns null
`parseByte_poorSignalCode`	BT Classic PoorSignal (code 0x02)
`signalQuality_*`	200/100/25/0 → NO_SIGNAL/POOR/FAIR/GOOD

Test location

```
sdk/src/test/kotlin/com/neurosky/sdk/parser/
└──── ThinkGearParserTest.kt
```

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16. API Reference

`NeuroSkySdk`

Main entry point. Manages BLE/BT Classic transport selection and lifecycle.

```
class NeuroSkySdk(context: Context)
```

Member	Type	Description
`connectionState`	`StateFlow<ConnectionState>`	Current connection state; hot Flow, always has a value
`dataFlow`	`Flow<BrainWaveData>`	Returns `activeTransport.dataFlow` at call time — collect **after** `connect()`
`connect(deviceAddress, transport)`	`suspend fun`	Connects via the given `TransportType`. Default: `TransportType.BLE`. No automatic

		fallback
`disconnect()`	`suspend fun`	Gracefully closes the active transport
`sendCommand(cmd: Byte)`	`suspend fun`	Sends a control byte to the headset
`findDeviceAddress(deviceName, timeoutMs)`	`suspend fun`	BLE scan returning MAC address, or null on timeout

> *`dataFlow` **timing:** `sdk.dataFlow` is a property getter — each access returns the current `activeTransport.dataFlow`. Capturing `sdk.dataFlow` into a variable before `connect()` will bind the flow to the pre-connect transport state. Always collect inside the same coroutine that calls `connect()`, or after `connectionState` emits `CONNECTED`.*
>
> *`TransportType` **not** `TransportMode`**:** The enum is `TransportType` with values `BLE` and `BT_CLASSIC`. `TransportMode` does not exist in the API.*

`BrainWaveData`

Immutable data class emitted by `dataFlow`.

Field	Type	Range	Description
`timestamp`	`Long`	—	`System.currentTimeMillis()` at receipt
`poorSignal`	`Int`	0~200	0 = perfect contact, 200 = no contact
`attention`	`Int`	0~100	eSense™ attention (0 = not computed)
`meditation`	`Int`	0~100	eSense™ meditation (0 = not computed)
`delta`	`Int`	0~∞	Delta band power, 0.5~2.75 Hz
`theta`	`Int`	0~∞	Theta band power, 3.5~6.75 Hz
`lowAlpha`	`Int`	0~∞	Low Alpha, 7.5~9.25 Hz
`highAlpha`	`Int`	0~∞	High Alpha, 10~11.75 Hz
`lowBeta`	`Int`	0~∞	Low Beta, 13~16.75 Hz
`highBeta`	`Int`	0~∞	High Beta, 18~29.75 Hz
`lowGamma`	`Int`	0~∞	Low Gamma, 31~39.75 Hz
`midGamma`	`Int`	0~∞	Mid Gamma, 41~49.75 Hz
`rawEeg`	`List<Int>`	-32768~32767	512Hz ADC samples (10 per packet)
`eyeBlink`	`Int`	0~255	Eye blink intensity; 0 = no blink
`signalQuality`	`SignalQuality`	enum	Derived from `poorSignal`

`ConnectionState`

```
enum class ConnectionState { DISCONNECTED, SCANNING, CONNECTING, CONNECTED, ERROR }
```

Value	Meaning
`DISCONNECTED`	No active connection
`SCANNING`	Scanning for the target device
`CONNECTING`	Device found, establishing connection
`CONNECTED`	Data stream active
`ERROR`	Connection attempt failed

`SignalQuality`

Derived from `BrainWaveData.poorSignal`.

Value	poorSignal	Reliability	Recommended action
`GOOD`	0	Excellent	Use all data
`FAIR`	1~50	Acceptable	Use data; minor noise present
`POOR`	51~199	Unreliable	Prompt user to adjust headset
`NO_SIGNAL`	200	No data	Prompt user to put on headset

`Transport` (interface)

Common interface implemented by `BleTransport`, `BtClassicTransport`, and `SimulatorTransport`.

```
interface Transport {
    val dataFlow:  Flow<BrainWaveData>
    val stateFlow: Flow<ConnectionState>
    suspend fun connect(deviceAddress: String)
    suspend fun disconnect()
    suspend fun sendCommand(cmd: Byte)
}
```

`SimulatorTransport`

```
class SimulatorTransport : Transport
```

Member	Description
`setMode(mode: Mode)`	Change simulation mode; takes effect on next emitted packet
`Mode.RANDOM`	Random Attention and Meditation values each tick

`Mode.FOCUSED`	Attention 70~~100, Meditation 40~~60
`Mode.RELAXED`	Attention 20~~50, Meditation 70~~100
`Mode.POOR_SIGNAL`	PoorSignal 150~200, Attention 0, Meditation 0

`NeuroSkyCommand`

```
object NeuroSkyCommand
```

Constant	Byte	When to use
`NOTCH_60HZ`	`0x1C`	Power grid is 60 Hz (Korea, USA)
`NOTCH_50HZ`	`0x1B`	Power grid is 50 Hz (Europe, China)
`START_RAW_EEG`	`0x15`	Enable raw EEG waveform (disabled by default)
`STOP_RAW_EEG`	`0x16`	Disable raw EEG waveform
`START_ESENSE`	`0x17`	Enable Attention/Meditation (enabled by default)
`STOP_ESENSE`	`0x18`	Disable Attention/Meditation

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